NEBULA GENERATOR

Single Point Interaction

Alex Pepper

Experimental Interaction - Summer ‘26

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Concept

A procedural nebula generator built in P5.js.
Each press of a single button produces a unique, unrepeatable image drawn from curated color palettes based on real Hubble Space Telescope nebula photography.

The project sits at the intersection of two personal interests: space and science fiction imagery. Rather than simulating a specific nebula, the generator produces something that feels scientifically plausible while remaining entirely abstract. A tool for making images that look like they could exist somewhere in the universe.

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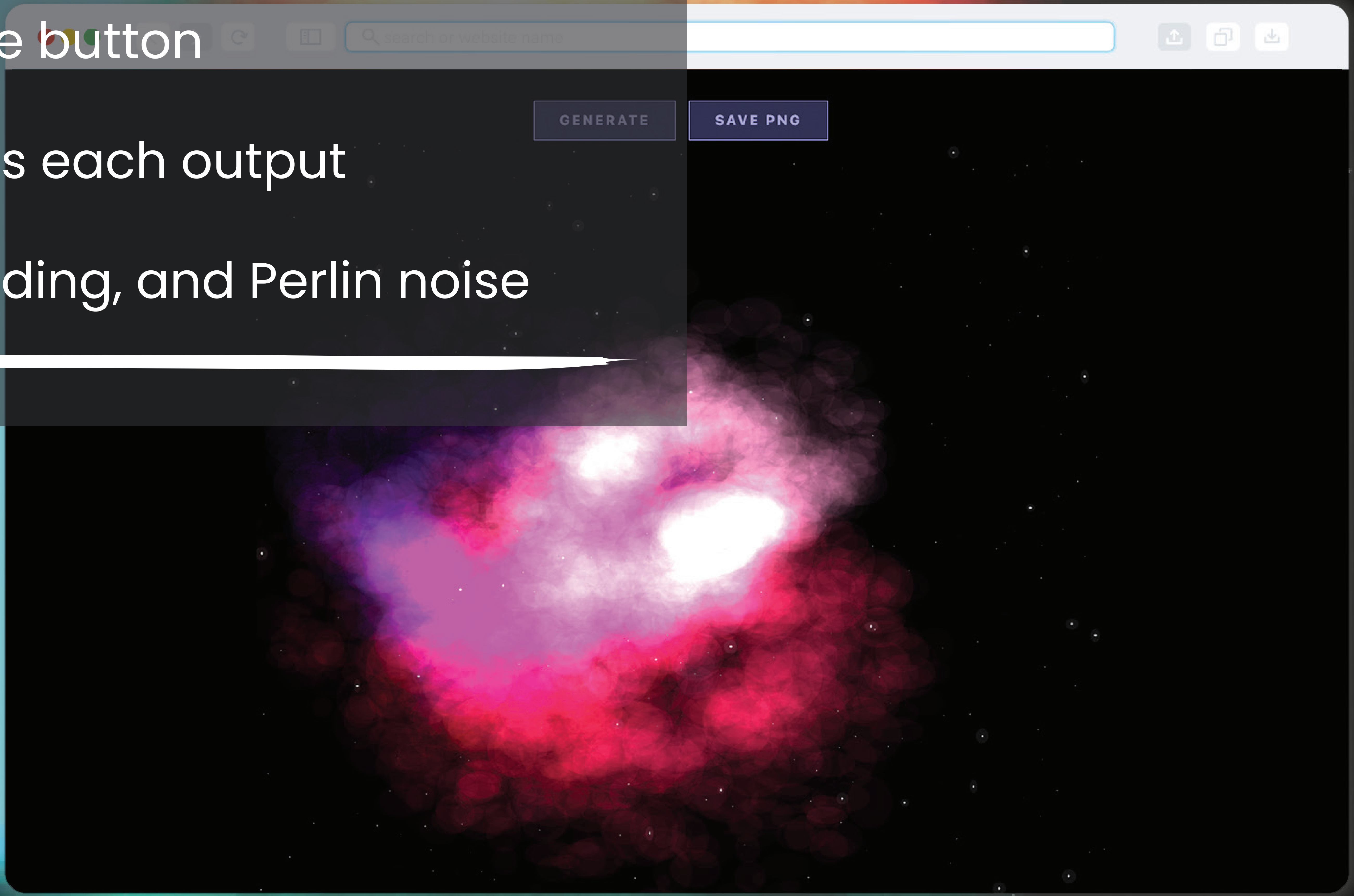
Draft Outputs

Single interaction: Generate button

Export: Save PNG downloads each output

Built with P5.js, additive blending, and Perlin noise



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Process + Next Steps

What's working:
Additive color blending produces convincing luminosity. Perlin noise shapes organic, non-repeating cloud forms. Palette system ensures every output is cohesive as a set.

What's next:
Refine palette range and cloud density. Explore adding a subtle title or designation to each output. Generate and curate the full set of 16 final outputs.

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